A library of extreme value theory in Lean 4

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Abstract

We describe a Lean 4 library, built on Mathlib, that formalizes the core of extreme value theory: the Fisher–Tippett–Gnedenko theorem, domains of attraction, regular variation, the Pickands–Balkema–de Haan theorem, estimators and likelihood theory for the tail, stable laws and heavy tails, extremes of dependent sequences, multivariate extremes and point processes. The library has about 100,000 lines and 4,300 theorems and lemmas, uses no `sorry` and depends only on Lean's three standard axioms, which continuous integration checks for every declaration. Since a formal proof does not show that a statement matches the literature, the library also carries specification tests, mutation checks and written audits, and it records where its statements are narrower than their textbook forms. Formalizing the extremal index exposed its hypotheses and led to examples that, to our knowledge, are new, including a process that answers a question of Bai, Kulik and Wang. The statements, tests and audits were written with AI assistance and have not yet been reviewed by human experts in extreme value theory.

1 Introduction

Extreme value theory describes the behaviour of maxima and of the far tail of a distribution. It is used to estimate quantities such as the level of a flood expected once in a hundred years or the loss from a catastrophic insurance event. Its results carry many technical hypotheses, and software such as `evd`, `ismev`, `extRemes` and `pyextremes` relies on them. A machine-checked account makes those hypotheses explicit.

This paper describes such an account, in Lean 4 [12] on top of Mathlib [25]. The development of the Fisher–Tippett–Gnedenko theorem and its blueprint are based on ExtremeValueProject, by Kalle Kytölä and the participants of an Aalto University course on formalized mathematics [21]. The source, an interactive blueprint, API documentation and a catalogue of results with their references are at `https://github.com/unbalancedparentheses/lean-extremevaluetheory`; this paper refers to release `v0.9.0` (doi:10.5281/zenodo.23966373).

2 Contributions

1. **Scope** (Sections 3 and 4). Within the limits of the search in Section 3, we know of no broader machine-checked development of extreme value theory, though the two central theorems have also been formalized independently in Isabelle. The library runs from the classical theory of maxima and regular variation through statistics to dependent sequences, multivariate extremes and point processes, on one set of conventions (Section 5).

*Repository and issue tracker: `https://github.com/unbalancedparentheses/lean-extremevaluetheory`.

2. **Testing statements, not only proofs** (Section 6). A proof assistant checks proofs. We describe the checks we use for the separate question of whether a statement is the right one, with their results and their limits.

3. **What the process found** (Sections 7 and 8): hypotheses in the literature that cannot be dropped, and examples for the extremal index that, to our knowledge, are new. These claims are provisional.

4. **An honest map** (Section 9): where the library is narrower than the textbooks, and what no one independent has yet checked.

3 Related work

We searched the web, GitHub and arXiv for formalizations of extreme value theory in Lean, Isabelle and Coq, and of regular variation, stable laws and the Skorokhod space. The search was not exhaustive: our first pass missed the Isabelle entries below, which a review of this draft found. So what follows reports what we found, not what exists. Table 1 compares the nearest work.

Extremes. Hirata's entries in the Archive of Formal Proofs [17, 18], dated September 2026, prove Fisher–Tippett–Gnedenko and Pickands–Balkema–de Haan in Isabelle, following de Haan–Ferreira and Embrechts et al. They are independent of this library, which makes them valuable. We compared the two formal statements of each theorem (`docs/HIRATA_COMPARISON.md`). By an argument we wrote, not a mechanized one, the statements of both main theorems are equivalent. His regular variation is more complete than ours in places, and we found no error in either. A note on the Extreme Value Theory entry says that more than 95% of its code was written by generative AI, so ours is not the only development made this way.

Our Fisher–Tippett–Gnedenko development and its blueprint are based on ExtremeValueProject [21], from which this library began. MathFin [7] is a Lean library of mathematical finance. It has about 900 lines on extremes, connected to its risk measures: GEV max-stability, concrete domains of attraction, GPD threshold stability and a Poisson–GPD–GEV identity. Its source states that neither Fisher–Tippett–Gnedenko nor Pickands–Balkema–de Haan is formalized there. Its aims complement ours, and its classification of each result by how faithful it is to the claim influenced our own plans.

Probability in proof assistants. This library depends on Mathlib's measure-theoretic probability [25, 32]. Avigad, Hölzl and Serafin proved the central limit theorem in Isabelle [1], and Degenne, Ledvinka, Marion and Pfaffelhuber constructed Brownian motion in Lean [10], which Coelho extended to an Itô calculus [8]. Our functional tail empirical process theorem does not use those developments: it constructs its limiting law on $D[0,1]$ and proves that it has Brownian finite-dimensional distributions, and related results use Mathlib's Brownian-motion API. We did not find formalizations of stable laws or the Skorokhod space.

The extremal index. The closest mathematics to Section 7 is the stable-regenerative model of Bai and Wang [3] and the analysis of Bai, Kulik and Wang [2].

	This work	Hirata [17, 18]	ExtremeValue-Project [21]	MathFin [24]
Prover	Lean 4	Isabelle/HOL	Lean 4	Lean 4
Focus	Extreme value theory	The two central theorems, domains of attraction, regular variation	A course project	Mathematical finance
Fisher–Tippett–Gnedenko	Proved	Proved, also for iid samples	The project's target; our development starts from it	Max-stability and concrete domains; no general classification
Domains of attraction	Gnedenko's conditions, quantile form, von Mises	Gnedenko's conditions, quantile forms, von Mises, canonical constants; normal, Erlang, uniform and Cauchy	–	Concrete domains
Peaks over threshold	Equivalence, every shape	Equivalence, every shape	–	GPD properties; not the theorem
Regular variation	Yes, with Breiman and multivariate	Uniform convergence, representation, Potter bounds, Karamata (direct and converse)	–	–
Statistics of extremes	Yes	–	–	–
Dependent sequences	Yes	–	–	–

Table 1: The nearest formalizations we found. A dash means we did not find this in the project. For Hirata's entries, this rests on reading their statements and definitions, not most of their proofs; `docs/HIRATA_COMPARISON.md` compares them. Entries describe the versions we read in November 2026.

4 What is formalized

The central result is the classification of limits of normalized maxima. Writing $A \bullet G$ for the distribution function G after an increasing affine change of variables, the library states it as follows.

```
theorem CumulativeDistributionFunction
    .IsExtremeValueDistr.classification
    {G : CumulativeDistributionFunction} (hG : G.IsExtremeValueDistr) :
    (∃ (A : AffineIncrEquiv), A • G = standardGumbelCDF) ∨
      (∃ (A : AffineIncrEquiv) (ξ : ℝ) (hξ : 0 < ξ),
        A • G = standardFrechetCDF hξ) ∨
      (∃ (A : AffineIncrEquiv) (ξ : ℝ) (hξ : 0 < ξ),
        A • G = standardWeibullCDF hξ)
```

Here `G.IsExtremeValueDistr` says that G is non-degenerate and is a limit of $F(a_n x + b_n)^n$ at continuity points of G, for some distribution function F and affine normalizations. This is the

theorem of Fisher and Tippett [15] and Gnedenko [16]. Table 2 summarizes the rest, following the standard accounts [13, 11, 6, 28, 23, 9]; the repository's `THEOREMS.md` lists each result with its source. Among the original sources are Balkema and de Haan [4] and Pickands [26] for peaks over threshold, Hill [19] for the tail estimator, Smith [29] for likelihood, Embrechts and Veraverbeke [14] for ruin, Leadbetter [22] for dependent sequences, and Basrak and Segers [5] for the tail process.

Area	Main results
Limits of maxima	Fisher–Tippett–Gnedenko; max-stable laws are exactly the extreme value laws
Domains of attraction	Fréchet, Weibull and Gumbel characterizations; the quantile form
Peaks over threshold	Pickands–Balkema–de Haan as an equivalence for every shape
Regular variation	Karamata (both halves), Potter bounds, representation theorem, Breiman's lemma
Stable laws	Generalized CLT for $0 < \alpha < 2$; domains of attraction; uniqueness of the index
Heavy tails	Subexponential laws, random sums, Embrechts–Veraverbeke ruin asymptotics
Estimators	Hill (consistency, normality, dependent data), Pickands and moment consistency, Weissman
Likelihood	GPD and GEV maximum likelihood: consistency, Fisher information; GPD joint normality
Return levels	Asymptotic normality, Wald intervals with asymptotic coverage, the delta method
Dependence	Leadbetter's theorem; extremal index; tail process; $\theta = \vartheta$ under anti-clustering and $A(u_n)$
Time series	AR(1), ARMAX, linear processes, Markov chains; $X = AX + B$ and GARCH(1,1) tail indices
Multivariate	Regular variation, converse Cramér–Wold, spectral representation, the multivariate GPD
Processes	Poisson random measures; exceedance point processes; $D[0,1]$ is Polish; tail empirical process

Table 2: Contents of the library. Section 9 qualifies these entries.

General results that are not specific to extremes (vague convergence, Poisson random measures, mixing coefficients, the Skorokhod space, the delta method) were absent from Mathlib when we needed them. They live in a separate folder, written to Mathlib's conventions so that they can be contributed upstream.

5 Design

One representation of laws. A probability law on $\mathbb{R}$ appears in three forms: a measure, which is the canonical law, as in Mathlib; Mathlib's distribution function of a measure; and `CumulativeDistributionFunction`, a Stieltjes function with limits 0 and 1, which carries the affine action, types and domains of attraction. `CDFBridge.lean` collects the bridges to Mathlib's distribution functions and laws, building on the equivalence with probability measures in `CumulativeDistributionFunction.lean`.

Conventions. Predicates are `Is...` structures or definitions with facts in their namespace, and names follow Mathlib's rules. Iid statistical results typically take measurable X_i with `iIndepFun` and either `IdentDistrib` or a specified common law. A theorem that is the counterpart of a textbook result says so in its docstring and cites the source. Renames leave a deprecated alias. The repository's `docs/DESIGN.md` records these rules.

Separation by purpose. General results go in `ForMathlib/` (31 files and about 8,400 lines), which imports only Mathlib and itself, to ease moving it upstream after review of naming, placement and overlap with Mathlib. Counterexamples showing that hypotheses are needed go in `Minimality/`. Tests of statements go in a separate `Test` library, which uses only public statements.

6 Testing the statements

Lean's kernel guarantees that a conclusion follows from its hypotheses. It does not guarantee that the hypotheses can hold, or that a definition means what its name suggests. In a formal development these failures are easy to produce: in Lean $x/0 = 0$, $\log 0 = 0$, and the integral of a non-integrable function is 0, so a statement can be true for an unintended reason. Several statements in the project this library started from were false as written and have been corrected. We use the following checks, all but the last of which can be rerun from the repository.

Axiom audit. Continuous integration fails if any of the library's 9,487 declarations depends on an axiom other than `propext`, `Classical.choice` and `Quot.sound`.

Independent kernel. A nightly job rechecks all declarations, including those of Mathlib, with `con-leche`, a second kernel.

Non-vacuity tests. 95 tests apply a headline theorem to a concrete object, showing that its hypotheses can hold together. They include the theorems named in Section 7.

Known-answer tests. 159 tests check definitions against textbook values, which catches sign and parametrization errors.

Junk-value audit. A scanner, not part of continuous integration, examined 3,718 theorems for statements that hold only through a totalization convention. Reviewing the 740 it flagged found one such statement, now fixed.

Mutation checks. A script, not part of continuous integration, tried 323 perturbations of the statements of 45 headline theorems with their proofs kept. A perturbation that still compiles suggests a theorem proved for a trivial reason. One survived, and we judged it harmless.

Faithfulness audits. Six written comparisons of statements with the literature, recording each deviation.

Table 3 gives the audits' verdicts on the 133 statements and definitions in their main table. These records can lag the code: the Codex review of this release (Section 11) found some scope notes that later theorems had overtaken.

These tests and audits were themselves written with AI assistance, so they are not independent of the statements they test. They reduce the risk of error, but do not remove it.

Verdict recorded in `FAITHFULNESS.md`	Count
Faithful: the statement matches the cited result	87
Faithful within a stated scope, such as a special case or an extra hypothesis	34
Differs from the cited form, but correct as stated	12

Table 3: Audit verdicts. They are the judgement of AI-assisted audits, not of experts.

7 What the formalization found

Hypotheses that cannot be dropped. The library contains counterexamples showing that hypotheses of Karamata's theorem, Breiman's lemma, the von Mises condition, the generalized CLT, Hill consistency, Leadbetter's theorem and the identification $\theta = \vartheta$ of the extremal index are needed.

The extremal index and the tail process. For a stationary, nonnegative, regularly varying sequence let θ be its extremal index, and let ϑ be the candidate extremal index read off the spectral tail process [20]. Then $\theta \le \vartheta$, and they agree when anticlustering and the block condition $A(u_n)$ hold on a common block sequence $r_n \to \infty$, $r_n = o(n)$. For signed sequences, the formalized comparison concerns $|X_t|$. Bai and Wang [3] gave a model with $\theta < \vartheta$. We give a much simpler one.

The *echo process* is built from iid Pareto(α) shocks Z_t and the lag $L(z) = \lfloor \log z \rfloor + 1$, by $X_t = \max\big(Z_t, \sup\{Z_s : s < t,\ s + L(Z_s) = t\}\big)$: each shock appears at its own time and again $L(Z_s)$ steps later. A large shock and its echo separate beyond every fixed window, so the spectral tail process is $\Theta_t = \mathbf{1}\{t = 0\}$ and $\vartheta = 1$. Yet each cluster of exceedances consists of a shock and its echo. Since $E[L(Z_0)] < \infty$, echoes entering from before the observation window are negligible, so $P(M_n \le u_n) \to e^{-\tau/2}$ whenever $nP(X_0 > u_n) \to \tau$, which gives $\theta = 1/2$ (`logEchoProcess_theta_lt_vartheta`). The library proves the same for other lags that grow without bound and have finite mean.

An answer to a question of Bai, Kulik and Wang. Bai, Kulik and Wang [2, arXiv v2, §1.4] ask whether there are models for which anticlustering holds with $\lim \log r_n / \log n = \rho$ only for ρ up to some $\rho_0 < 1$. Given $\rho_0 \in (0, 1)$, take the lag $\lfloor z^\gamma \rfloor + 1$ with $\gamma = \alpha\rho_0$. For $0 < \rho < 1$ with $\rho \ne \rho_0$, anticlustering at the normalizing levels holds if and only if $\rho < \rho_0$ (`powEchoProcess_antiClustering_iff`). The formal statement uses one level for both the conditioning event and the exceedance, where their condition allows two. The two forms are equivalent by an elementary argument from regular variation, which is not formalized.

Mixing and a warning for estimators. The echo processes are β-mixing, geometrically for the logarithmic lag, and still have $\theta \ne \vartheta$ (`logEchoProcess_betaMixing_incompatible`). This illustrates the classical point that anticlustering and mixing must hold at a common block length. On the logarithmic-lag process, write $p_n = P(X_0 > u_n)$ for thresholds $u_n \to \infty$. If $np_n \to \infty$, $r_n \to \infty$, $r_n = o(\log u_n)$, and there are gaps $s_n \to \infty$ with $\log n = o(s_n)$ and $r_n + s_n = o(np_n)$, then the runs estimator converges in probability to $1 = \vartheta$, not to $\theta = 1/2$ (`logEcho_tendstoInMeasure_runsEstimator`).

Status of these claims. Our literature check compared these results with the papers we could read in full. We could not obtain the Kulik–Soulier book [20], Weissman–Cohen [31], or Smith (1988) [30], so the claim of novelty is provisional, and no expert has confirmed it. Some results close to these are known: $\theta \le \vartheta$ is Lemma 7.5.4 of [20] as cited by [2], and the principle behind the mixing example is classical.

8 A proof in detail: the echo process

The echo results take about 4,400 lines of Lean, including the gap formula below. For a measurable lag $L \ge 1$ with $L(z) \to \infty$ and $E[L(Z_0)] < \infty$, the argument runs as follows.

Sources of an exceedance. For $x \ge 1$, almost surely $X_t > x$ holds exactly when some shock is its source: either $Z_t > x$, or $Z_{t-j} > x$ with $L(Z_{t-j}) = j$ for some $j \ge 1$ (`echoSource`). Borel–Cantelli ensures that only finitely many shocks echo into each time. Summing over j gives the upper bound $P(X_0 > x) \le P(Z_0 > x) + \sum_{j\ge1} P(Z_{-j} > x,\, L(Z_{-j}) = j) = 2P(Z_0 > x)$. For the lower bound we need that these events rarely overlap, and a pairwise-independence inequality $P(\bigcup_j A_j) \ge S - S^2$, where $S = \sum_j P(A_j)$ (`le_measureReal_iUnion_of_pairwise`), supplies it. So $P(X_0 > x) \sim 2x^{-\alpha}$ (`tendsto_measureReal_lt_echoProcess_div`).

The tail process. Conditionally on $X_0 > x$, the source is the shock at 0, whose echo is at lag $L(Z_0) \to \infty$, or a shock at lag $-L \to -\infty$. At any fixed lag $t \ne 0$, nothing else is large. A general criterion, that this asymptotic independence forces $\Theta_t = \mathbf{1}\{t = 0\}$, then gives $\vartheta = 1$ (`hasSpectralTailProcess_echoProcess`). This step uses $L \to \infty$ but not the finite mean.

The maxima. Write $M_n = \max_{0\le k<n} X_k$. Outside the event that an above-threshold shock from before time 0 echoes into the window, $M_n \le u_n$ holds exactly when $Z_0, \dots, Z_{n-1} \le u_n$. Finite mean makes those echoes negligible. If $nP(X_0 > u_n) \to \tau$ then $nP(Z_0 > u_n) \to \tau/2$, so $P(M_n \le u_n) \to e^{-\tau/2}$ and $\theta = 1/2$ (`hasExtremalIndex_echoProcess`). The computation of θ is direct, and does not use the block condition.

The gap. A general theorem explains the discrepancy. Let $T_n = \inf\{t \ge 1 : |X_t| > u_n\}$, and define the *late-return mass* η as the iterated limit, in n and then m, of $E[(1 - T_n/r_n)^+ \mathbf{1}\{T_n > m\} \mid |X_0| > u_n]$. For a stationary series with a spectral tail process and $u_n, r_n \to \infty$, the cluster ratio tends to ℓ exactly when $\eta = \vartheta - \ell$, with no anticlustering or mixing assumed (`HasSpectralTailProcess.tendsto_clusterRatio_iff`). If also $nP(|X_0| > u_n) \to \tau > 0$, $r_n = o(n)$ and $A(u_n)$ holds, then $\theta = \vartheta - \eta$, so that $\theta < \vartheta$ exactly when $\eta > 0$ (`HasSpectralTailProcess.lt_candidateExtremalIndex_iff`). For the echo process, $\eta = 1/2$ when $r_n u_n^{-\alpha} \to 0$ and $E[L(Z_0); Z_0 > u_n] = o(r_n u_n^{-\alpha})$ (`hasLateReturnMass_echoProcess`). This is close to a reformulation of classical blocks-and-runs identities. We did not find the two-sided statement in the literature.

What the formalization contributed. Each lemma states the hypotheses it uses, which is how we can say that $\vartheta = 1$ needs only $L \to \infty$, and that $\theta \le \vartheta$ needs no mixing or anticlustering. The formal counterexample also corrected our own understanding: notes we wrote earlier had read a lemma of Planinić and Soulier [27] as saying that $|Y_t| \to 0$ could replace anticlustering in

$\theta = \vartheta$. The echo process has $|Y_t| \to 0$ and $\theta < \vartheta$, so that reading is false. The error was ours, and we do not claim one in their paper.

9 Limitations

Many results are narrower than their textbook forms. The README and `FAITHFULNESS.md` give the full list; the most important are the following.

- **No expert review.** This matters most, and the rest of this list may be incomplete because of it.
- **GEV likelihood.** Maximum likelihood consistency is proved on a compact parameter set and for an exact iid GEV sample, not for block maxima from a law in a domain of attraction. Joint normality of the three-parameter estimator is not proved, so GEV and Gumbel return-level normality assume parameter normality as a hypothesis.
- **GPD likelihood and return levels.** Joint normality needs $\xi_0 > -1/2$, $\xi_0 \neq 0$ and a compact parameter set. Peaks-over-threshold return levels treat the excesses as an exact iid GPD sample with a known exceedance probability.
- **Estimators.** Normality of the Pickands and moment estimators is not proved, and neither are profile-likelihood intervals. The blocks and runs estimators of the extremal index assume that their population ratio converges to θ, and the intervals estimator assumes its empirical-moment limits.
- **Ruin.** The ruin probability is defined by the Pollaczek–Khinchine formula, not derived from the risk process.
- **Stochastic recurrence.** The tail of $X = AX + B$ is identified through moment bounds. The sharp Kesten–Goldie asymptotics is not proved.
- **Dependence.** Several theorems assume noise that is nonnegative at every sample point.
- **Process limits.** Point-process convergence is formalized through joint laws of counts on finite families of disjoint sets, with no topology on the space of measures. The functional limit of the tail empirical process is for uniform observations in unweighted $D[0, 1]$; beyond it, only weighted near-zero bounds are proved.
- **Asymptotics only.** The library says nothing about finite-sample behaviour or about choosing a threshold or block length.

10 Evaluation and reproducibility

Evidence. The library has about 100,000 lines and 4,300 theorems and lemmas, in 9,487 declarations counted by the axiom audit. On the latest run before this release, the independent kernel accepted all 719,515 declarations of the library and its dependencies. A build that passes in continuous integration is itself the evidence for the axiom claim.

To reproduce. Release `v0.9.0` (doi:10.5281/zenodo.23966373) pins Lean v4.35.0 and Mathlib v4.35.0 (commit `56a00585`) in `lean-toolchain` and `lake-manifest.json`. Then:

```
git clone https://github.com/unbalancedparentheses/lean-extremevaluetheory
cd lean-extremevaluetheory && git checkout v0.9.0
lake exe cache get     # prebuilt Mathlib
lake build             # the library
lake build Test        # specification tests and the axiom audit
```

A weekly job builds a fresh project that requires the repository and imports only the umbrella module, which checks that others can use it as a dependency.

Upstream. Nothing has been contributed to Mathlib yet. `UPSTREAMING.md` plans how the `ForMathlib/` files could move there as a series of small pull requests.

11 How it was built

Most of the code, tests and documents were written by AI coding agents (Claude Code), directed and reviewed by the author, on top of the Aalto project. Codex, a second AI system, reviewed the statements and claims of this release. This changes what evidence the library can offer. The proofs need no trust in the agents, because the kernel checks them. The statements do, and that is the reason for Section 6 and for the request below.

12 Conclusion

We would most value review by extreme value theorists. The repository's `docs/REVIEW.md` translates the main statements into ordinary mathematics, so no knowledge of Lean is needed, and corrections are very welcome. It does not yet cover the answer to Bai, Kulik and Wang's question, return levels, confidence intervals or the extremal-index estimators; `FAITHFULNESS.md` lists those. Next come the missing likelihood results, contributing the general results to Mathlib, and executable estimators cross-checked against R and Python packages.

What this library is not is part of the point. It is not software for analysing data, it says nothing about finite samples, and it does not replace statistical judgement about thresholds, block lengths or whether a model fits. It provides a checked account of what the asymptotic theory says and under exactly which hypotheses.

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